

ThmSmith Stage -1 proposal: **flagship_explore** / **f2**

Assignment-memory mixture splitting for dynamic IV bounds under imperfect compliance

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Locked parameters. This is a partial-identification anchor, not a panel pool-mode. The locked tuple is

$$(\text{qid}, \text{specialization}, P, \theta, \text{identifying restrictions}, \text{bounding functional}) \\ = (\text{flagship_explore}, \text{f2}, P_{Y, D_1, D_2, Z_1, Z_2, X}, \theta_{C,0}^{(2)}, \mathcal{R}_{\text{2stageIV+memory+1sided}}, [\theta_{\text{mem}}(P, e), \bar{\theta}_{\text{mem}}(P, e)]).$$

Here X has finite support, $Z_t, D_t \in \{0, 1\}$, and $Y \in [0, 1]$. The first-stage encouragement Z_1 is randomized within X ; the second-stage encouragement follows the known adaptive law

$$e_2(z \mid x, z_1, d_1) = \mathbb{P}(Z_2 = z \mid X = x, Z_1 = z_1, D_1 = d_1),$$

which is allowed to depend on the retained assignment-memory label Z_1 even after conditioning on (X, D_1) . The target is the first-stage-complier, first-stage-untreated second-period effect

$$\theta_{C,0}^{(2)} = \mathbb{E}[Y(0, 1) - Y(0, 0) \mid D_1(1) = 1, D_1(0) = 0].$$

The restrictions $\mathcal{R}_{\text{2stageIV+memory+1sided}}$ are consistency, exclusion of Z_1, Z_2 except through D_1, D_2 , randomized Z_1 within X , sequential randomization of Z_2 given (X, Z_1, D_1) , positivity of all reachable encouragement cells, one-sided first-stage compliance $D_1(0) = 0 \leq D_1(1)$, one-sided second-stage compliance within each retained history, and a no-direct-memory restriction equating $Y(0, d_2)$ and second-stage response laws across $(Z_1, D_1) = (0, 0)$ and $(1, 0)$ within the same first-stage principal stratum. The bounding functional is the sharp interval obtained by a finite latent-response linear program that retains Z_1 as an assignment-memory splitter inside the $D_1 = 0$ state.

Abstract

Covariate-adaptive and sequentially randomized experiments with imperfect compliance are usually analyzed either as static IV bounds within baseline strata or as dynamic point-identification problems under strong cross-period conditions. This proposal asks whether a previous randomized encouragement can tighten partial-identification bounds after it has ceased to be a treatment-relevant state variable. The kernel conjecture is that, in a two-stage binary design, retaining Z_1 inside the $D_1 = 0$ branch yields a strict mixture-splitting gain for the second-period effect among first-stage compliers. The gain comes from a compliance-revealing pure never-taker cell, not from adding another baseline covariate or from a Bellman recursion. If resolved, the result would give a small, auditable partial-identification theorem explaining when adaptive assignment memory has identifying content under imperfect compliance.

1 Introduction

Why the problem is important. Sequential experiments often adapt later encouragements to earlier assignment and treatment history. When subjects do not perfectly comply, the earlier randomized encouragement is more than an administrative label: under one-sided first-stage compliance, $Z_1 = 1, D_1 = 0$ reveals a never-taker stratum, while $Z_1 = 0, D_1 = 0$ is a mixture of never-takers and latent first-stage compliers. The methodological question is whether this randomized label should be retained when bounding later dynamic treatment effects.

The main finding (proposed). Theorem 1 records the routine but important information-ordering result: the sharp memory-retaining identified set is contained in the identified set obtained after coarsening away Z_1 inside the $D_1 = 0$ state. Conjecture 1 is the flagship statement: for the first-stage-complier second-period effect $\theta_{C,0}^{(2)}$, the full-memory set equals a covariate-cell mixture-subtraction envelope that combines the mixed $Z_1 = 0, D_1 = 0$ branch with the pure-never-taker $Z_1 = 1, D_1 = 0$ branch. Conjecture 2 states that this envelope is strictly narrower than the coarsened/static envelope whenever the pure-never-taker second-stage IV problem is informative and the first-stage complier share is nondegenerate.

What is new. The closest published paper is Bugni et al. (2024), which characterizes ATE and ATT identified sets under covariate-adaptive ran-

domization and imperfect compliance. Their assignment strata are pre-treatment covariates; the proposed gap is that a post-baseline randomized encouragement can become a principal-stratum splitter after conditioning on treatment receipt. The closest dynamic point-identification paper is [Sojitra and Syrgkanis \(2024\)](#), which identifies dynamic LATEs under one-sided noncompliance plus extra cross-period conditions; the proposed gap is what remains partially identified when those extra conditions are not imposed. Unlike the burned `flagship_explore/f1` angle, this proposal does not claim a Bellman-versus-full-history separation for a sustained regime effect; it isolates a two-stage mixture-subtraction face created by one pure compliance stratum.

How it positions in the literature. The proposal lives in the partial-identification cluster, specifically IV bounds with dynamic treatment timing and imperfect compliance. It uses Balke–Pearl finite response-type logic for the second-period IV subproblems, the covariate-adaptive randomization motivation of recent imperfect-compliance work, and the principal-stratum language of SMART analyses. The intended contribution is a theorem-level diagnostic for when assignment memory sharpens a dynamic partial-ID interval, not a new graphical identification rule or a panel regression fact.

2 Related work

[Balke and Pearl \(1997\)](#) derive tight nonparametric IV bounds for static binary randomized encouragement with imperfect compliance; the present proposal uses the same finite latent-response philosophy inside each second-stage history. [Swanson et al. \(2018\)](#) review binary-IV partial identification and organize the static assumptions that the proposed dynamic cell problem reduces to when the first-stage memory label is removed. [Manski and Pepper \(2000\)](#) show how monotone instrumental-variable and response restrictions can sharpen static bounds; this proposal is orthogonal because the sharpening comes from randomized mixture splitting rather than an outcome shape restriction.

[Bugni et al. \(2024\)](#) characterize ATE and ATT identified sets under covariate-adaptive randomization and imperfect compliance; this proposal asks whether an adaptive post-baseline assignment label has additional identifying content after conditioning on earlier receipt. [Bugni, Canay, and Shaikh \(2018\)](#) study inference under covariate-adaptive randomization without making dynamic compliance states the estimand, providing the randomization-

design background. [Sojitra and Syrgkanis \(2024\)](#) identify dynamic LATEs under one-sided noncompliance and additional cross-period effect-compliance independence; the current proposal weakens the target to bounds and studies the mixture information left by Z_1 . [Michael et al. \(2024\)](#) identify marginal structural mean model parameters with time-varying IVs under compliance-type restrictions; this proposal instead stays nonparametric and records a finite sharp set. [Chen and Zhang \(2023\)](#) develop IV-optimal dynamic treatment regimes through Bellman-style partial identification; this proposal avoids the burned Bellman kernel and focuses on a single second-period principal-stratum contrast. [Artman et al. \(2024\)](#) model partial compliance in SMARTs using principal-strata marginal structural models and Bayesian semiparametrics; the proposed theorem asks what a nonparametric bounds analysis can recover before imposing that model. [Murphy \(2005\)](#) supplies the SMART and dynamic-regime design context in which retained assignment history is operationally natural.

In-catalogue prior art is sparse for this anchor. The `doc/API.md` ExactID/PartialID catalogue reports that ThmSmith partial-identification entries are empty at seed time, with promoted partial-ID work organized around static Manski-family and Balke–Pearl-family bounds. The closest banked proposal, `pid_late_bounded_defiers`: “Sharp complier-LATE bounds from a collapsed Balke–Pearl principal-stratum envelope,” is one-period and does not use a post-baseline randomized memory label. The burned `flagship_explore/f1`: “Full-history IV bounds for dynamic treatment effects under adaptive noncompliance” targeted sustained-regime full-history LP separation; the present proposal changes the kernel to a two-stage compliance-revealing mixture split for $\theta_{C,0}^{(2)}$.

3 Formal setup

Let X have finite support and let $Z_t, D_t \in \{0, 1\}$ for $t = 1, 2$. The observed law is $P = P_{Y, D_1, D_2, Z_1, Z_2, X}$, with $Y \in [0, 1]$. First-stage potential treatment is $D_1(z_1)$, and the first-stage principal stratum is

$$S = C \text{ if } D_1(1) = 1, D_1(0) = 0, \quad S = N \text{ if } D_1(1) = 0, D_1(0) = 0.$$

For x , write $q_C(x) = \mathbb{P}(S = C \mid X = x)$ and $q_N(x) = 1 - q_C(x)$. Under one-sided first-stage compliance and randomized Z_1 , $q_C(x) = \mathbb{P}(D_1 = 1 \mid Z_1 = 1, X = x)$, while $Z_1 = 1, D_1 = 0$ is a pure N cell and $Z_1 = 0, D_1 = 0$ is a $q_C(x)$ -by- $q_N(x)$ mixture.

Second-stage potential treatment is $D_2^{s,d_1}(z_2)$ for first-stage stratum $s \in \{C, N\}$ and first-stage treatment $d_1 \in \{0, 1\}$. Second-stage potential outcomes are $Y(d_1, d_2)$. Define the cell-level target

$$\theta_{C,0}^{(2)}(x) = \mathbb{E}[Y(0, 1) - Y(0, 0) \mid S = C, X = x],$$

and the aggregate target

$$\theta_{C,0}^{(2)} = \sum_x w_C(x) \theta_{C,0}^{(2)}(x), \quad w_C(x) = \frac{\mathbb{P}(X = x) q_C(x)}{\sum_{x'} \mathbb{P}(X = x') q_C(x')}.$$

For each x and $s \in \{C, N\}$, let

$$\mathcal{B}_{s,0,x}(d_2) \subseteq [0, 1]$$

denote the sharp second-stage IV identified set for $\mu_{s,d_2}(x) = \mathbb{E}[Y(0, d_2) \mid S = s, X = x]$, using the relevant observed history and the second-stage one-sided IV restrictions. For $s = N$, this set is informed by the pure cell $(Z_1, D_1) = (1, 0)$. Let $\mathcal{M}_{0,x}(d_2)$ denote the sharp second-stage IV identified set for the mixed branch mean

$$m_{0,d_2}(x) = q_C(x) \mu_{C,d_2}(x) + q_N(x) \mu_{N,d_2}(x)$$

inside $(Z_1, D_1) = (0, 0)$. The full-memory identified set $\Theta_{\text{mem}}(P, e)$ is the set of all values of $\theta_{C,0}^{(2)}$ generated by a finite latent-response distribution satisfying the assumptions and reproducing the full observed law P . The coarsened identified set $\Theta_{\text{coarse}}(AP, e)$ uses the same latent variables and target but replaces P by the law AP that sums over Z_1 within the $D_1 = 0$ branch after conditioning on X .

```
qid=flagship_explore
specialization=f2
P=P_{Y,D1,D2,Z1,Z2,X}, a finite-support observed law with Y in [0,1]
theta=theta_C0^(2)=E[Y(0,1)-Y(0,0) | D1(1)=1,D1(0)=0]
identifying restrictions=R_2stageIV+memory+1sided: consistency,
    exclusion, randomized Z1 within X, sequentially randomized Z2 given
    (X,Z1,D1), positivity, one-sided first- and second-stage compliance, and
    no direct assignment-memory effect within the D1=0 state
bounding functional=[theta_mem_lower(P,e), theta_mem_upper(P,e)], the sharp
    interval over the finite latent first-stage stratum, second-stage response,
    and bounded outcome table retaining Z1 as a mixture-splitting label
```

4 Assumptions

Assumption 1 (Finite support and bounded outcomes). X has finite support; $Z_1, Z_2, D_1, D_2 \in \{0, 1\}$; and $0 \leq Y(d_1, d_2) \leq 1$ almost surely for all $d_1, d_2 \in \{0, 1\}$.

Assumption 2 (First-stage random encouragement). For every x used in the target, $0 < \mathbb{P}(Z_1 = 1 \mid X = x) < 1$, and Z_1 is independent of $(D_1(0), D_1(1), D_2^{s,d_1}(0), D_2^{s,d_1}(1), Y(d_1, d_2))$ conditional on $X = x$.

Assumption 3 (Sequential second-stage random encouragement). For every reachable (x, z_1, d_1) , the known assignment probability $e_2(z \mid x, z_1, d_1)$ lies strictly between zero and one, and Z_2 is independent of the latent first-stage stratum, second-stage response type, and potential outcomes conditional on $(X, Z_1, D_1) = (x, z_1, d_1)$.

Assumption 4 (Consistency and exclusion). The realized treatments satisfy $D_1 = D_1(Z_1)$ and $D_2 = D_2^{S,D_1}(Z_2)$. The realized outcome is $Y = Y(D_1, D_2)$, and Z_1, Z_2 affect Y only through D_1, D_2 .

Assumption 5 (One-sided compliance). First-stage compliance is one-sided: $D_1(0) = 0 \leq D_1(1)$ almost surely. For each s, d_1 , second-stage compliance is one-sided: $D_2^{s,d_1}(0) = 0 \leq D_2^{s,d_1}(1)$ almost surely.

Assumption 6 (Nondegenerate first-stage mixture). For every x used in the strict-gain claim, $0 < q_C(x) < 1$. Boundary cells with $q_C(x) = 0$ are excluded from $\theta_{C,0}^{(2)}$, and cells with $q_N(x) = 0$ are treated by the sanity reduction in Corollary 1.

Assumption 7 (No direct assignment-memory effect in the untreated first-stage state). Within $X = x$, first-stage stratum s , and first-stage treatment $D_1 = 0$, the second-stage response law $D_2^{s,0}(z_2)$ and the potential outcomes $Y(0, d_2)$ do not depend directly on whether the retained label was $Z_1 = 0$ or $Z_1 = 1$.

Assumption 8 (Informative pure never-taker branch). For at least one x , the pure-never-taker second-stage IV set $\mathcal{B}_{N,0,x}(d_2)$ is a proper subset of $[0, 1]$ for $d_2 = 0$ or $d_2 = 1$, and the mixed-branch set $\mathcal{M}_{0,x}(d_2)$ is compatible with a nonempty subtraction interval for $\mu_{C,d_2}(x)$.

5 Main results

Theorem 1 (Memory refinement is never weaker). *Under Assumptions 1–7,*

$$\Theta_{\text{mem}}(P, e) \subseteq \Theta_{\text{coarse}}(AP, e).$$

Equivalently,

$$\underline{\theta}_{\text{coarse}}(AP, e) \leq \underline{\theta}_{\text{mem}}(P, e) \leq \bar{\theta}_{\text{mem}}(P, e) \leq \bar{\theta}_{\text{coarse}}(AP, e).$$

Why this is new and worth studying. This theorem is mathematically routine, but it fixes the ordering for the new object. Bugni et al. (2024) compare identified sets under covariate-adaptive randomization for static ATE/ATT targets; here the refinement is a post-baseline assignment-memory refinement for a principal-stratum dynamic effect, so the theorem identifies the right null benchmark for the conjectures.

Conjecture 1 (Mixture-subtraction envelope). *(NOT YET PROVED.) Under Assumptions 1–7, fix x with $0 < q_C(x) < 1$. For each $d_2 \in \{0, 1\}$, the sharp full-memory cell identified set for $\mu_{C,d_2}(x)$ is*

$$\mathcal{I}_{C,x}^{\text{mem}}(d_2) = \left\{ \frac{m - q_N(x)n}{q_C(x)} : m \in \mathcal{M}_{0,x}(d_2), n \in \mathcal{B}_{N,0,x}(d_2), 0 \leq \frac{m - q_N(x)n}{q_C(x)} \leq 1 \right\}.$$

Consequently, the sharp full-memory cell identified set for $\theta_{C,0}^{(2)}(x)$ is the interval difference

$$\mathcal{I}_{C,x}^{\text{mem}}(1) - \mathcal{I}_{C,x}^{\text{mem}}(0).$$

Why this is new and worth studying. The closest static result is Balke and Pearl (1997); the closest CAR result is Bugni et al. (2024). Neither isolates the algebraic face where a randomized prior assignment reveals one component of a later imperfect-compliance mixture, so resolving this conjecture would convert a generic two-period latent LP into a closed-form diagnostic.

Conjecture 2 (Strict assignment-memory gain). *(NOT YET PROVED.) Under Assumptions 1–8, there exists an open set of observed laws P and adaptive laws e_2 with*

$$e_2(1 \mid x, 0, 0) \neq e_2(1 \mid x, 1, 0)$$

for which the containment in Theorem 1 is strict. In those laws, the strictness is witnessed by at least one endpoint of $\mathcal{I}_{C,x}^{\text{mem}}(d_2)$ being pulled inward by the proper pure-never-taker set $\mathcal{B}_{N,0,x}(d_2)$. If $q_N(x) = 0$, or if $\mathcal{B}_{N,0,x}(0) = \mathcal{B}_{N,0,x}(1) = [0, 1]$ for every x , this particular mixture-splitting mechanism cannot yield strict improvement.

Why this is new and worth studying. Sojitra and Syrgkanis (2024) obtain point identification by adding cross-period independence-style conditions, while Michael et al. (2024) identify model parameters under compliance-type restrictions. This conjecture proposes a different phenomenon: even without point identification, a remembered randomization label can strictly sharpen a nonparametric bounds problem.

Conjecture 3 (Covariate aggregation by complier weights). *(NOT YET PROVED.) Under Assumptions 1–7, if the latent restrictions are imposed separately within each $X = x$ cell and no cross- x shape restriction is added, then the sharp aggregate set for $\theta_{C,0}^{(2)}$ is the weighted Minkowski sum*

$$\Theta_{\text{mem}}(P, e) = \sum_x w_C(x) (\mathcal{I}_{C,x}^{\text{mem}}(1) - \mathcal{I}_{C,x}^{\text{mem}}(0)) .$$

Why this is new and worth studying. Swanson et al. (2018) emphasize the value of putting IV bounds in common notation for static problems. A proved aggregation statement would make the proposed memory-split theorem usable in stratified SMART or covariate-adaptive designs without hiding cross-cell assumptions.

6 Sanity-check corollaries

Corollary 1 (No mixture, no memory gain). *Under Assumptions 1–7, if $q_N(x) = 0$ for every target cell, then $(Z_1, D_1) = (0, 0)$ contains only first-stage compliers and the mixture-subtraction envelope collapses to the ordinary second-stage one-sided IV bound for that group.*

Under Assumptions 1–7, the statement collapses because $m_{0,d_2}(x) = \mu_{C,d_2}(x)$ when $q_N(x) = 0$, so subtracting the pure-never-taker component removes a zero-mass term.

Corollary 2 (Perfect second-stage compliance reduces to principal-stratum subtraction). *Add to Assumptions 1–7 the restriction $D_2^{s,0}(z_2) = z_2$ for*

$s \in \{C, N\}$. Then $\mathcal{B}_{N,0,x}(d_2)$ and $\mathcal{M}_{0,x}(d_2)$ are observed conditional-mean singletons, and Conjecture 1 reduces to the standard two-component mixture identity

$$\mu_{C,d_2}(x) = \frac{m_{0,d_2}(x) - q_N(x)\mu_{N,d_2}(x)}{q_C(x)}.$$

Under Assumptions 1–7 plus perfect second-stage compliance, the result is a direct algebraic reduction: the second-stage IV subproblem has no hidden compliance type, so the only remaining hidden object is the first-stage C/N mixture.

7 Proof-sketch route

The proof route is finite-dimensional convex analysis. Encode each $X = x$ cell by a latent table containing $S \in \{C, N\}$, the second-stage one-sided response types under $D_1 = 0$, and bounded potential-outcome coordinates $Y(0, 0), Y(0, 1)$. Theorem 1 follows by applying the linear coarsening map A to the full observed-cell equations. For Conjecture 1, eliminate the pure-never-taker variables using the $(Z_1, D_1) = (1, 0)$ equations, then eliminate the mixed-branch mean equations in $(Z_1, D_1) = (0, 0)$; the expected output is a projection of a small polytope onto $(\mu_{C,0}, \mu_{C,1})$ whose endpoints are exactly the displayed subtraction envelope. The strict-gain claim should be checked first on a binary-outcome, single- x , $q_C = q_N = 1/2$ witness by enumerating the second-stage response types and comparing the two LP endpoint systems with and without the Z_1 label.

8 Honest scope flags

Theorem 1 is expected to be routine once the two feasible sets are defined. Conjectures 1, 2, and 3 are not yet proved; the main risk is that second-stage imperfect compliance may introduce cross-history compatibility constraints not captured by the displayed one-dimensional subtraction form. The no-direct-memory assumption is strong and should be treated as part of the estimand convention, not as a harmless technicality. This pivot deliberately withdraws the burned f1 claims about Bellman non-sharpness, sustained-regime full-history separation, and minimality in T .

9 Iteration trail

- v1 (pivot) – initial draft for a structurally different angle: assignment-memory mixture splitting for $\theta_{C,0}^{(2)}$; prior verdict: none.

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